

Eduardo Alejandro Aviles Jimenez

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Machine Learning Engineer with 9+ years of experience spanning full-stack development, data engineering, and AI/ML solutions across financial services, social media, and enterprise applications. Expert in building end-to-end ML pipelines, deep learning, NLP, computer vision, and predictive analytics. Proven leader in architecting scalable cloud infrastructure and DevOps practices that reduce operational costs by 30%. Strong background in technical innovation and business impact, delivering production-ready AI solutions with regulatory compliance.

EDUCATION

Computer Systems Engineering — *Engineering Degree*
Universidad De San Carlos De Guatemala

JAN 2005 - SEPT 2011

WORK EXPERIENCE

LoanPro — *Machine Learning Engineer*

JUN 2022 - PRESENT

- Implemented a Loan Creation Monitor, cluster user behavior, and identify anomalies, improving tenant targeting.
- Developed an anomaly detection system that reduced tenant noise data by 25%, filtering irrelevant events allowing the business to focus on critical insights.
- Created per-tenant smart threshold calculation, considering data distribution, multiple modes, skewness, and kurtosis, optimizing sensitivity for better decision-making.
- Built automated model workflows with continuous training (CT) and model validation to monitor online model performance, preventing model degradation.
- Developed a custom data pipeline for training and inference, using ECS and Fargate for scalability and reliability.
- Implemented a custom LLM AI assistant to summarize and analyze loan data and provide insights and recommendations.
- Developed Risk Radar project using **XGBoost** to predict loan delinquency across 3-month windows, creating comprehensive risk scores that align with IFRS 9 regulatory requirements for Expected Credit Loss (ECL) provisioning.
- Built predictive models calculating probability-weighted risk scores across three delinquency stages (Current, Early/Severe Delinquency, Default), enabling proactive risk management and reducing potential losses by identifying high-risk loans before default.
- Created Risk Radar dashboard with tenant risk leaderboard and early warning alerts, improving portfolio oversight and enabling data-driven decision making for loan servicing teams.

BairesDev - HipTrain — *Senior Software Engineer*

AUG 2021 - JUN 2022

- Led infrastructure, architecture, and CI/CD pipelines for back-end and front-end solutions, improving deployment times by 30%.
- Developed scheduling and reservation core system for training classes, improving scheduling by 80%.
- Built custom generic calendar invite logic for scheduling, improving integration and user convenience.
- Implemented a chatbot using Amazon Lex, which automated 40% of customer inquiries and improved user interaction efficiency.
- Led and managed back-end and infrastructure projects, ensuring scalable architecture and seamless integration with existing systems.

LogMeIn — *BI/ML Developer*

APR 2016 - DEC 2021

- Developed infrastructure for data integration and ETL with multiple sources, improving data processing capabilities.
- Built microservices for custom ETL solutions, enhancing scalability and reliability.
- Implemented ML algorithms for product enhancements, leading to 20% increase in prediction accuracy.

SSNAPP (Social Media Gateways) — *Senior Software Developer*

SEP 2014 - APR 2016

- Led transition from monolithic architecture to microservices using Netflix OSS, increasing system scalability by 50% and reducing downtime by 20%.
- Optimized back-end infrastructure and implemented AWS solutions, reducing infrastructure costs by 30%.

SKILLS

- **Programming Languages:** Python, Java, TypeScript, JavaScript, SQL
- **Web & Backend Frameworks:** Node.js, React, SolidJS, NestJS, Spring Boot, FastAPI, Django, Flask
- **Machine Learning & AI:** XGBoost, Scikit-Learn, TensorFlow, Keras, MLFlow, Pandas, DVC, Weights& Biases, LlamaIndex
- **Cloud & Infrastructure:** AWS (ECS, Fargate, Lambda), Docker, Kubernetes, Terraform, Jenkins, CloudFormation, Ansible
- **Databases & Storage:** PostgreSQL, Redis, RethinkDB, DynamoDB, Redshift
- **Financial & Compliance:** IFRS 9, Expected Credit Loss (ECL), Regulatory Reporting, Risk Modeling
- **Development Tools:** Git, GitHub Actions, RabbitMQ, Netflix OSS, Amazon Lex

AWARDS AND CERTIFICATES

- Hackathon/LoanPro - Risk Radar — *First Place*** - JAN 2025
- Risk Radar: Advanced loan delinquency prediction system using XGBoost with IFRS 9 compliance, featuring probability-weighted risk scoring and automated portfolio monitoring dashboard.
- Hackathon/LoanPro - Loan Monitoring — *First Place*** - APR 2024
- Loan monitoring system that utilized clustering and anomaly detection techniques to improve tenant targeting.
- Hackathon/LogMeIn — *First Place*** - JUL 2018
- Created an LSTM-based neural network to predict customer churn, leveraging advanced recurrent neural network architectures to solve critical business problems related to customer retention. Also voted as developers' choice.
- Nano Degree — *Machine Learning DevOps Engineer*** JAN 2022 - APR 2022
- Production-ready ML pipelines, automated workflows with CI/CD, FastAPI, MLFlow, Weights&Biases, and model performance tracking.
- Nano Degree — *Cloud DevOps Engineer*** JUL 2019 - AUG 2020
- Infrastructure as code (IaC) with CloudFormation and Ansible, CI/CD pipelines with Jenkins, and Kubernetes microservices.
- Nano Degree — *Artificial Intelligence Nanodegree*** JAN 2018 - JUL 2018
- AI algorithms, search optimization, planning, and pattern recognition.
- Nano Degree — *Deep Learning Nanodegree*** JUL 2017 - AUG 2017
- Image classification, sentiment analysis, and face generation using CNNs, RNNs, and GANs.

TECHNICAL PROJECTS

- LoanPro — 🏠 *Risk Radar - Loan Delinquency Prediction System*** JAN 2025 - PRESENT
- Developed comprehensive loan delinquency prediction system using **XGBoost** models to calculate probability-weighted risk scores across three IFRS 9-compliant stages: Current/Performing, Early/Severe Delinquency, and Default/Charged-off.
 - Implemented sophisticated risk scoring algorithm that considers multiple delinquency probabilities over 3-month windows, enabling proactive risk management and early intervention strategies.
 - Built Risk Radar dashboard with tenant risk leaderboard, automated alerts, and drill-down capabilities, providing actionable insights for loan servicing teams and portfolio managers.
- LoanPro — 🏠 *Loan Monitoring System*** JAN 2024 - DEC 2024
- Designed and deployed a system to monitor loan creation, cluster user behavior, and identify anomalies, using **MLFlow**, **Python**, **Pandas**, and **Scikit-learn**. Structured data pipeline for Training and Inference with **ECS** and **Fargate** for scalability and reliability.
- Personal Project — 🤖 *Predict Customer Churn*** JAN 2024 - APR 2024
- Python package for customer churn prediction, adhering to PEP8 standards with best practices for modularity, documentation, and testing.
- Personal Project — 🤖 *ML Pipeline for Rental Price Prediction*** JAN 2024 - APR 2024
- End-to-end ML pipeline for NYC rental price prediction with automated weekly retraining and deployment.
- Personal Project — 🤖 *Dynamic Risk Assessment System*** JAN 2024 - APR 2024
- Automated ML model lifecycle management with data ingestion, scoring, retraining, and deployment.

SPEAKING ENGAGEMENTS

- FaceBook dev Circles, Guatemala — *Neural Network from scratch*** - MARCH 2019
- Implemented neural network without external ML libraries, showcasing fundamental AI concepts with raw Python.
- FaceBook dev Circles, Guatemala — *RNNs and Sequence to Sequence Model*** - OCT 2018
- Custom RNN-based model to generate fake social media threads, trained on "Velocidad Maxima" dataset.
- Mayan Linguistic Conference, Guatemala — *Neural Machine Translation*** - JUL 2018
- Text translation evolution and neural models for Mayan dialect translation and language preservation.
- Google IO Extended, Guatemala — *Generative Adversarial Networks (GAN)*** - MAY 2018
- Keynote on GAN implementation for facial image generation, demonstrating GAN versatility across applications.